

10-Mark Questions & Answers — Data Structures Module 1: ADT, Arrays & Pointers

Q1. Explain Abstract Data Types (ADT), 1D/2D/3D Array Memory Representations, Row/Column Major Address Calculations, and Pointers in detail with C programs.

Comprehensive 10-mark master answer covering:

1. **Concept of ADT:** Definition, separation of Interface vs Implementation, axioms, and examples (List ADT, Stack ADT).
2. **Types of Data Structures:** Primitive vs Non-Primitive, Linear (Array, Linked List, Stack, Queue) vs Non-Linear (Tree, Graph), Static vs Dynamic.
3. **Address Calculation Formulas:**
 - **1D Array:** $\text{Address}(A[i]) = \text{Base} + w * (i - \text{LowerBound})$
 - **2D Row-Major:** $\text{Address}(A[i][j]) = \text{Base} + w * [(i - \text{LBR}) * N_{\text{cols}} + (j - \text{LBC})]$
 - **2D Column-Major:** $\text{Address}(A[i][j]) = \text{Base} + w * [(j - \text{LBC}) * N_{\text{rows}} + (i - \text{LBR})]$
4. **Pointers & Multidimensional Arrays:** Dynamic allocation with `malloc()` and array of pointers in C.
5. **Self-Referential Structures:** Defining linked nodes using `struct Node { int data; struct Node *next; };`.